

ECN 594: Identification and Instrumental Variables

Nicholas Vreugdenhil

Plan for today

1. **Recap: Logit and Berry inversion**
2. The identification problem
3. Direction of bias
4. Uber example
5. Instrumental variables
6. Elasticities and limitations

Recap: Logit and Berry inversion

- Mean utility: $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}$ ($\alpha < 0$)
- Logit shares:

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^J \exp(\delta_{kt})}, \quad s_{0t} = \frac{1}{1 + \sum_{k=1}^J \exp(\delta_{kt})}$$

- **Berry inversion** (from last time):

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}$$

- This looks like a regression we can run... but there's a problem
- **Today:** Price endogeneity and instrumental variables

Plan for today

1. Recap: Logit and Berry inversion
2. **The identification problem**
3. Direction of bias
4. Uber example
5. Instrumental variables
6. Elasticities and limitations

The estimating equation (from Lecture 1)

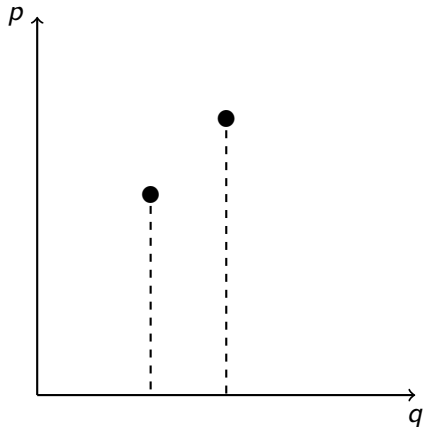
- From Berry inversion:

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} \quad (\alpha < 0)$$

- This looks like a regression we can run with OLS
- **But there's a problem...**

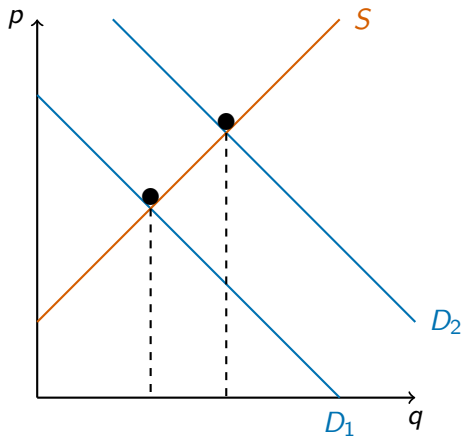
The classic identification problem

- We observe equilibrium prices and quantities
- Question: what's the demand curve?



The classic identification problem

- We observe equilibrium prices and quantities
- Problem: cannot tell if demand or supply is shifting



The classic identification problem

- If we just regress q on p , what do we get?
- Neither demand nor supply!
- **Question:** What variation do we need to identify demand?

The classic identification problem

- If we just regress q on p , what do we get?
- Neither demand nor supply!
- **Question:** What variation do we need to identify demand?
- **Key insight:** We need variation that shifts ONE curve but not the other
 - To identify demand: need **supply shifters**
 - To identify supply: need **demand shifters**

Price endogeneity in demand estimation

- Our estimating equation:

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} \quad (\alpha < 0)$$

- ξ_{jt} = unobserved product quality
- **Problem:** Firms observe ξ_{jt} when setting prices!
- High quality products (ξ_{jt} high) tend to have high prices
- $\Rightarrow \text{Cov}(p_{jt}, \xi_{jt}) > 0$
- OLS gives biased estimates

Plan for today

1. Recap: Logit and Berry inversion
2. The identification problem
3. **Direction of bias**
4. Uber example
5. Instrumental variables
6. Elasticities and limitations

Direction of bias

- If high-quality products have high prices...
- OLS sees: high price, but demand still high (because of ξ)
- OLS concludes: price doesn't hurt demand much
- **Result:** $\hat{\alpha}$ biased toward zero (less negative than truth)

Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5 .
- Is this consistent with endogeneity bias? Why?

Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5 .
- Is this consistent with endogeneity bias? Why?
- **Answer:** Yes!
 - OLS overstates how much consumers like expensive products
 - So OLS finds a smaller (less negative) price coefficient
 - $-0.3 > -0.5$, so this is exactly what we'd expect

Plan for today

1. Recap: Logit and Berry inversion
2. The identification problem
3. Direction of bias
4. **Uber example**
5. Instrumental variables
6. Elasticities and limitations

Case study: Uber surge pricing

- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**

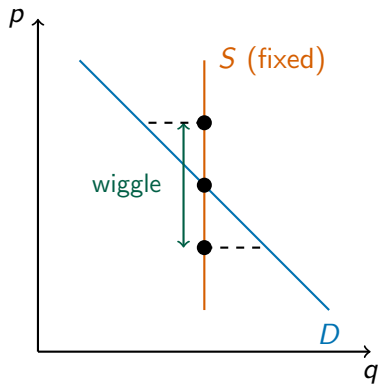
Case study: Uber surge pricing

- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**
- Surge responds to BOTH:
 - **Supply shocks:** Fewer drivers available → surge up
 - **Demand shocks:** Concert ends, rain starts → surge up
- Can't just regress rides on surge multiplier!
- Same problem as before: observing equilibrium prices

The solution: Uber's price “wiggle” experiment

- Uber's insight: **randomize** prices!
- **Cohen et al. (2016)**: Uber experimentally varied surge multipliers
- Same time, same location → different riders see different prices
- Key: Some riders randomly see 1.9x, others see 2.1x
- Demand conditions are identical, only price differs

Tracing out the demand curve



- Randomization holds supply fixed
- Price “wiggles” up/down randomly
- Traces out points ON the demand curve
- This identifies demand!

Uber experiment: Results

- Result: demand elasticity ≈ -0.5
- **Inelastic!** A 10% price increase $\rightarrow$ only 5% fewer rides
- Why so inelastic?
 - When you need a ride, you *need* a ride
 - Limited substitutes late at night
- **Key lesson:** Experiments give clean identification
- But most industries can't randomize prices...

Why experiments are powerful

- No confounding: price variation is independent of demand shocks
- Clean identification of the demand curve
- **But:**
 - Tech companies can run experiments
 - Traditional industries can't randomize prices
 - Most IO settings require **instrumental variables**

Plan for today

1. Recap: Logit and Berry inversion
2. The identification problem
3. Direction of bias
4. Uber example
5. **Instrumental variables**
6. Elasticities and limitations

Instrumental variables: the solution

- Need variables z that are:
 1. **Relevant:** Correlated with price ($\text{Cov}(z, p) \neq 0$)
 2. **Exogenous:** Uncorrelated with ξ ($\text{Cov}(z, \xi) = 0$)
- These are cost shifters or other supply-side variables

Common IVs in demand estimation

Three main approaches used in the literature:

1. **Cost shifters:** Input prices, exchange rates
2. **Hausman IVs:** Prices in other markets
3. **BLP IVs:** Characteristics of competing products

Let's examine each in detail...

IV #1: Cost shifters

- **Idea:** Use variables that shift marginal costs
 - Input prices (steel, labor, energy)
 - Exchange rates (for imported goods)
 - Tariffs, taxes, regulations

IV #1: Cost shifters

- **Idea:** Use variables that shift marginal costs
 - Input prices (steel, labor, energy)
 - Exchange rates (for imported goods)
 - Tariffs, taxes, regulations
- **Relevance:** Higher costs $\rightarrow$ higher prices
- **Exogeneity:** Cost shocks don't directly affect consumer utility
 - Steel prices don't make you like/dislike a car

IV #1: Cost shifters

- **Idea:** Use variables that shift marginal costs
 - Input prices (steel, labor, energy)
 - Exchange rates (for imported goods)
 - Tariffs, taxes, regulations
- **Relevance:** Higher costs $\rightarrow$ higher prices
- **Exogeneity:** Cost shocks don't directly affect consumer utility
 - Steel prices don't make you like/dislike a car
- **Problem:** Cost data rarely observed!
 - Firms don't disclose detailed cost information
 - Even aggregate input prices may be hard to match to products

IV #2: Hausman instruments

- **Idea:** Use prices of the same product in *other markets*
 - Coca-Cola price in Chicago as IV for Coca-Cola price in Phoenix

IV #2: Hausman instruments

- **Idea:** Use prices of the same product in *other markets*
 - Coca-Cola price in Chicago as IV for Coca-Cola price in Phoenix
- **Why it works:**
 - Same product has similar costs across markets
 - Cost shocks (ingredients, shipping) affect prices everywhere
- **Key assumption:** Demand shocks ζ_{jt} are independent across markets
 - Chicago's unobserved demand factors don't affect Phoenix

Worked example: When do Hausman IVs fail?

- **Question:** You want to use Pepsi prices in Denver as an IV for Pepsi prices in Phoenix. When might this fail?

Worked example: When do Hausman IVs fail?

- **Question:** You want to use Pepsi prices in Denver as an IV for Pepsi prices in Phoenix. When might this fail?
- **Answer:**
- Hausman IVs assume $\tilde{\zeta}_{jt}$ is independent across markets
- **This fails when:**
 - National advertising campaigns affect all markets
 - Nationwide product reformulation ("New Coke")
 - Coordinated promotions or seasonal demand patterns
- **Relevance is OK:** Cost shocks do affect prices everywhere
- **Exogeneity is the problem:** Demand shocks may also be correlated!

IV #3: BLP instruments

- **Idea:** Use characteristics of *other products* in the market
- **Key assumption:**

$$E(\xi_{jt}|x_t) = 0$$

“Observed characteristics are mean independent of unobserved characteristics”

- **Common BLP instruments:**
 1. Own product characteristics (not price!)
 2. Sum of characteristics of *own-firm* products
 3. Sum of characteristics of *competitor* products
- **Intuition:** More/closer competitors $\rightarrow$ more competition $\rightarrow$ lower prices

BLP instruments: Example

- Consider estimating demand for the Honda Civic
- **BLP instruments could include:**
 - Civic's own characteristics: horsepower, MPG, size
 - Sum of Honda's other cars' characteristics (Accord, CR-V)
 - Sum of competitors' characteristics (Corolla, Mazda3, etc.)

BLP instruments: Example

- Consider estimating demand for the Honda Civic
- **BLP instruments could include:**
 - Civic's own characteristics: horsepower, MPG, size
 - Sum of Honda's other cars' characteristics (Accord, CR-V)
 - Sum of competitors' characteristics (Corolla, Mazda3, etc.)
- **Why exogenous?**
 - Toyota's decision about Corolla's horsepower doesn't affect Civic's unobserved quality ξ
- **Why relevant?**
 - If Toyota improves Corolla, Honda faces more competition
 - This pressures Honda to lower Civic's price

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?
- **Relevance:** Higher gas prices $\rightarrow$ higher operating costs $\rightarrow$ might affect car prices? Maybe weakly.
- **Exogeneity:** Do gas prices affect car quality ζ_{jt} ?
 - Gas prices affect *demand* for fuel-efficient cars
 - This might shift which cars look “good” to consumers
 - Potentially problematic!
- **Verdict:** Probably not a great IV

Summary: IV conditions

- For z to be a valid IV:
 1. **Relevant:** z must predict prices
 - Can test this! Run first-stage regression
 2. **Exogenous:** z must not affect demand directly
 - Cannot test this directly (requires economic reasoning)
- This is the standard IV framework from econometrics
- Applied to demand estimation: use supply-side variation

What makes a good IV? Summary

Instrument	Relevance	Exogeneity
Cost shifters (input prices)	Strong	Usually OK
Hausman IVs (prices elsewhere)	Strong	Debatable
BLP IVs (competitor chars)	Medium	Usually OK
Experiments (Uber wiggle)	Perfect	Perfect

- No perfect instrument in observational data
- **Best practice:** Use multiple IVs, check robustness
- **Experiments** are gold standard but often infeasible

Plan for today

1. Recap: Logit and Berry inversion
2. The identification problem
3. Direction of bias
4. Uber example
5. Instrumental variables
6. **Elasticities and limitations**

From estimation to elasticities

- We've seen how to estimate demand with logit
- **Why do we care?**
- Demand estimates let us compute **elasticities**
 - How sensitive are consumers to price changes?
 - How do consumers substitute between products?
- These are crucial for policy: mergers, pricing, welfare
- **But:** the logit model has important limitations...

Logit elasticities

- Given shares $s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_k \exp(\delta_{kt})}$
- We can derive price elasticities:

$$\eta_{jj} = \frac{\partial s_{jt}}{\partial p_{jt}} \frac{p_{jt}}{s_{jt}} = \alpha p_{jt} (1 - s_{jt}) \quad (\text{own-price})$$

$$\eta_{jk} = \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}} = -\alpha p_{kt} s_{kt} \quad (\text{cross-price})$$

- Note: $\alpha < 0$, so own-price elasticity is **negative** (as expected)

Limitation #1: Own-price elasticity

- Own-price elasticity: $\eta_{jj} = \alpha p_{jt}(1 - s_{jt})$
- With many products: $(1 - s_{jt}) \approx 1$
- So: $\eta_{jj} \approx \alpha p_{jt}$
- **Implication:** Own-price elasticity is proportional to price!

Limitation #1: Own-price elasticity

- Own-price elasticity: $\eta_{jj} = \alpha p_{jt}(1 - s_{jt})$
- With many products: $(1 - s_{jt}) \approx 1$
- So: $\eta_{jj} \approx \alpha p_{jt}$
- **Implication:** Own-price elasticity is proportional to price!
- Lower-priced products have more *inelastic* demand
- Higher-priced products have more *elastic* demand
- **Question:** Is this realistic?
 - Generic cereal (low price) should be more elastic than organic cereal?
 - Budget airline (low price) more inelastic than first class?

Limitation #2: Cross-price elasticity (IIA)

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k

Limitation #2: Cross-price elasticity (IIA)

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k
- Example: BMW raises its price
 - Logit says: same fraction go to Mercedes as to Honda Civic!
 - Reality: BMW consumers mostly switch to Mercedes, Audi
- This is the **IIA** (Independence of Irrelevant Alternatives) property

The IIA problem: Diversion ratios

- When product k 's price rises, where do consumers go?
- **Diversion ratio** from k to j :

$$D_{k \rightarrow j} = \frac{\partial s_{jt} / \partial p_{kt}}{\partial s_{kt} / \partial p_{kt}} = \frac{s_{jt}}{1 - s_{kt}}$$

- Substitution is proportional to **market share**, not similarity!

The IIA problem: Diversion ratios

- When product k 's price rises, where do consumers go?
- **Diversion ratio** from k to j :

$$D_{k \rightarrow j} = \frac{\partial s_{jt} / \partial p_{kt}}{\partial s_{kt} / \partial p_{kt}} = \frac{s_{jt}}{1 - s_{kt}}$$

- Substitution is proportional to **market share**, not similarity!
- **Why this matters:**
 - Merger analysis: which products are close substitutes?
 - Product positioning: who are my real competitors?
 - Logit can't answer these questions well

What causes the IIA problem?

- IIA comes from the **logit error structure**
- Key assumption: ε_{ijt} are *independent* across products
- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar

What causes the IIA problem?

- IIA comes from the **logit error structure**
- Key assumption: ε_{ijt} are *independent* across products
- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar
- **Solution:** Allow for consumer heterogeneity
 - High-income consumers prefer both BMW and Mercedes
 - When BMW's price rises, these consumers switch to Mercedes

Key Points

1. **Price is endogenous:** Firms observe ξ_{jt} when pricing $\rightarrow$ OLS biased toward zero
2. **Solution:** Instrumental variables (cost shifters, BLP IVs, Hausman IVs)
3. **Experiments** (like Uber) provide cleanest identification
4. **Logit elasticities:** $\eta_{jj} = \alpha p_{jt}(1 - s_{jt})$, $\eta_{jk} = -\alpha p_{kt}s_{kt}$
5. **IIA problem:** Cross-elasticity doesn't depend on $j \Rightarrow$ unrealistic substitution